

Geospatial Data Analysis

Identifying High-Potential Underserved Regional Markets

Dataset note: No regional dataset was provided for this task. Therefore, a synthetic sales dataset was created solely to demonstrate the requested geospatial analysis workflow. The results are illustrative and should not be treated as real company performance data.

1. Objective

The objective is to visualize location-based sales activity, compare demand and revenue across regions, and identify geographic markets that may have expansion potential.

2. Methodology

The dataset contains State, City, Postal Code, latitude/longitude, monthly users, and revenue. Duplicate records and invalid numeric values were checked and removed. Records were aggregated at city level. Demand was represented by total users, while revenue efficiency was measured as revenue per user. A Potential Score was calculated as 65% normalized demand plus 35% normalized coverage gap. Only markets at or above the 40th percentile of total users were considered for the final underserved-market ranking, preventing very small markets from dominating the result.

3. Dataset Summary

The synthetic dataset contains **120** transaction-month records across **30** cities and **21** states. Total simulated revenue is approximately **■16.63 million** and total simulated users are **8,422**.

Region	Users	Revenue	Revenue/User	Potential Score
Chennai, Tamil Nadu	436	■1,031,604	■2,366	0.716
Kolkata, West Bengal	413	■910,236	■2,204	0.708
Bengaluru, Karnataka	460	■1,211,299	■2,633	0.705

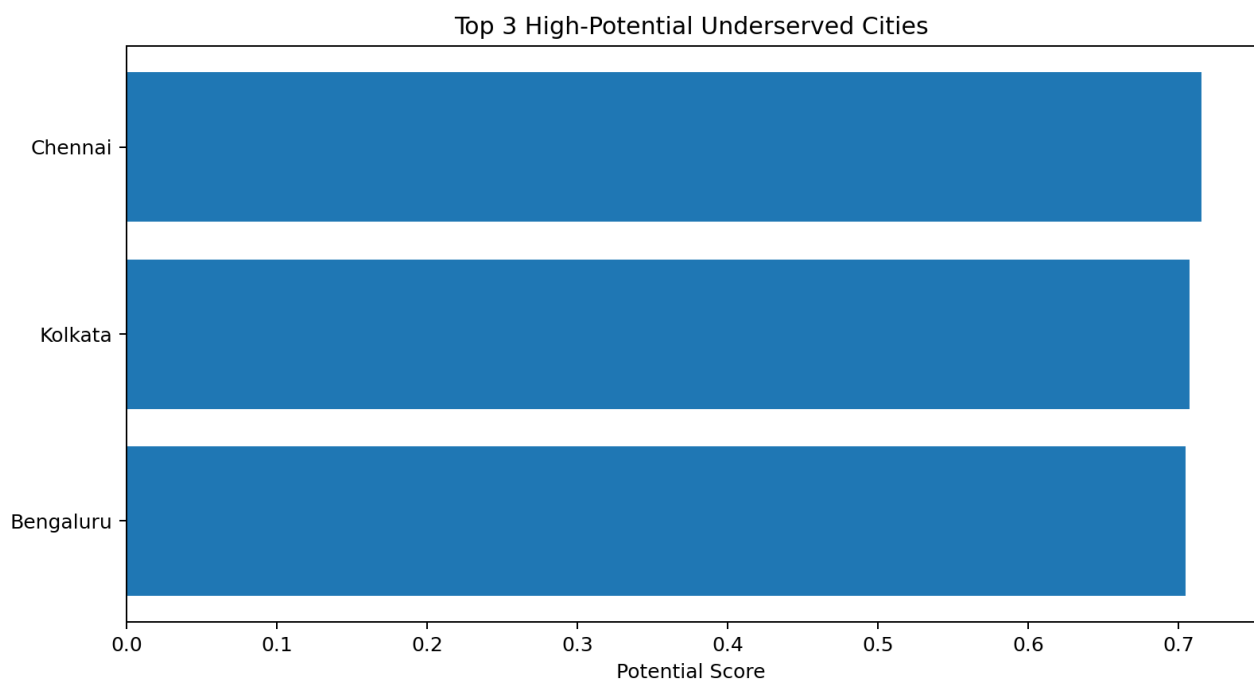
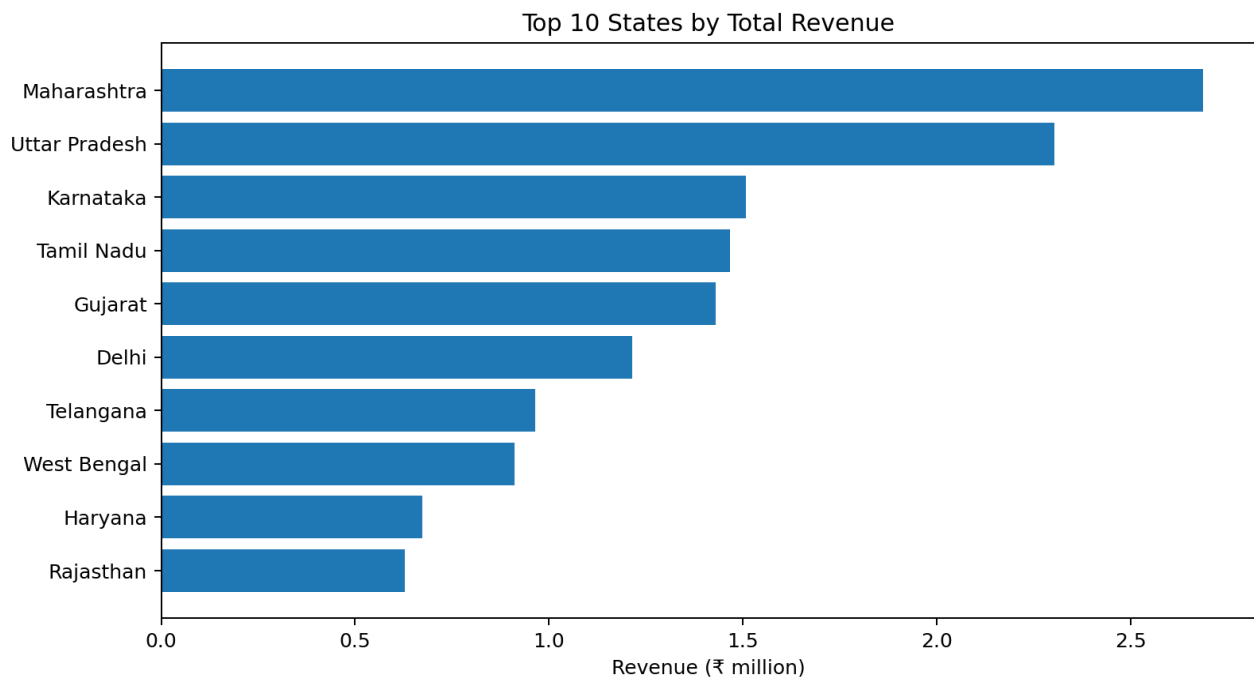
4. Top 3 High-Potential Underserved Regions

1. Chennai, Tamil Nadu — 436 users, ■1,031,604 revenue, revenue per user ■2,366, potential score 0.716. This combination indicates comparatively strong demand with lower revenue efficiency/coverage than stronger markets.

2. Kolkata, West Bengal — 413 users, ■910,236 revenue, revenue per user ■2,204, potential score 0.708. This combination indicates comparatively strong demand with lower revenue efficiency/coverage than stronger markets.

3. Bengaluru, Karnataka — 460 users, ■1,211,299 revenue, revenue per user ■2,633, potential score 0.705. This combination indicates comparatively strong demand with lower revenue efficiency/coverage than stronger markets.

5. Visual Analysis



6. Geographic Interpretation

The interactive Folium map places each analyzed city at its geographic coordinates. Larger highlighted markers identify the three highest-scoring expansion opportunities. Clicking a marker displays users, revenue, revenue per user, and the calculated potential score. This makes it possible to compare geographic demand with commercial performance.

7. Recommendations

1. Prioritize pilot expansion or localized campaigns in the top three identified markets. 2. Investigate why revenue efficiency is lower despite strong user demand; possible causes include limited product availability, weaker distribution, or lower conversion. 3. Use the interactive map to compare nearby markets before selecting a final expansion location. 4. Replace the synthetic dataset with verified company/regional data before making

real business decisions.

8. Conclusion

The analysis demonstrates how sales, demand, and geographic information can be combined to detect potentially underserved markets. The workflow is reproducible in Python and Folium and can be extended to state-level choropleths, real postal-code boundaries, or Power BI/Tableau once an official dataset is available.